KIIT Deemed to be University
Online End Semester Examination(Autumn Semester-2020)

Subject Name & Code: Data Structure and Algorithms CS-2001
Applicable to Courses: B.Tech

Full Marks=50 **Time:2 Hours**

SECTION-A(Answer All Questions. Each question carries 2 Marks)
Time:30 Minutes **(7×2=14 Marks)**

Question No	Question Type(MCQ/SAT)	Question	CO Mapping	Answer Key (For MCQ Questions only)

Q.No:1	MCQ	What is the time complexity of the following function func( )
		
		void func( )
		{
		int i, j;
		for (i=1; i<=n; i++)
		for(j=1;j<=log(i);j++)
		printf("KIIT University");
		}
		
		a) O(n)
		b) $O(n^2)$
		c) O(nlogn)
		d) O(logn)
			CO2	(c)

	MCQ	What is the time complexity of the following function fun( )
		void fun(int m, int n)
		{
		int i,j;
		int c =2;
		for (i = 1;i<=m; ++i) {
		for (j = 1;j<=m; ++j)
		printf("[%d,%d]", i,j);
		}
		for (i = 1;i<=n; ++i) {
		for (j = 1;i<=m; i=i*c)
		printf("[%d,%d], i,j);
			CO2	c)

		} **}** **a) O(n^2)** **b) O(m^2+mlogn)** **c) O(m^2+nlogm)** **d) O(m^2+n^2)**		
	MCQ	**What is the time complexity of the following function fun( )** **void fun(int m, int n)** **{** **int i,j;** **int c =2;** **for (i = 1;i<=n; ++i) {** **for (j = 1;j<=n; ++j)** **printf("[%d,%d]", i,j);** **}** **for (i = 1;i<=n; ++i) {** **for (j = 1;i<=m; i=i*c)** **printf("[%d,%d], i,j);** **}** **}** **a) O(n^2)** **b) O(n^2+mlogn)** **c) O(n^2+nlogm)** **d) O(m^2+n^2)**	CO2	c)
	MCQ	**What is the time complexity of the following function func( )** **void func( )** **{** **int i, j;** **for (i=1; i<=n; i=i+10)** **printf("KIIT University");** **}** **a) O(n)** **b) O(n^2)** **c) O(nlogn)** **d) O(logn)**	CO2	a)
Q.No:2	**MCQ**	The following function reverse() is suppose d to reverse a singly linked list. There is one line missing at the end of the functio n.What should be added in place of "/*AD D A STATEMENT HERE*/", so that the	CO3	a)

		function correctly reverses a linked list.
		
		static void reverse(struct node** head_ref)
		{
		struct node* prev = NULL;
		struct node* current = *head_ref;
		struct node* next;
		while (current != NULL)
		{
		next = current->next;
		current->next = prev;
		current = next;
		}
		/*ADD A STATEMENT HERE*/
		}
		a) *head_ref = prev;
		b) *head_ref = current;
		c) *head_ref = next;
		d) *head_ref = NULL;		

	MCQ	The function is called with the list contain ing the integers 1, 2, 3, 4, 5, 6, 7 in the given order. What will be the contents of the list after the function completes execut ion?
		
		void rearrange(struct node *list)
		{
		struct node *p, * q;
		int temp;
		if ((!list) || !list->next)
		return;
		p = list;
		q = list->next;
		while(q)
		{
		temp = p->value;
		p->value = q->value;
		q->value = temp;
		p = q->next;
		q = p?p->next:0;
		}
		}
		a) 1, 2, 3, 4, 5, 6, 7
		b) 2, 1, 4, 3, 6, 5, 7
		c) 1, 3, 2, 5, 4, 7, 6
		d) 2, 3, 4, 5, 6, 7, 1
			CO3	b)

	MCQ	Consider the function f defined here:
		struct item
		{
		int data;
		struct item * next;
		};
			CO3	b)

		int f (struct item *p)
		{
		return((p==NULL)
		||((p->next==NULL)||(p->data<=p->next->data) && (p->next)));
		}
		
		For a given linked list p, the function f returns 1 if and only if
		a) the list is empty or has exactly one element
		b) the element in the list are sorted in non-decreasing order of data value
		c) the element in the list are sorted in non-increasing order of data value
		d) not all element in the list have the same data value		

	MCQ	The following C function takes a singly linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code left blank.
		
		Node* move_to_front(Node* head)
		{
		Node* p, *q;
		if((head==NULL)| (head->next==NULL))
		return head;
		q=NULL;
		p=head;
		while(p->next != NULL)
		{
		q=p;
		p=p->next;
		}
		
		_____
		return head;
		}
		
		Choose the correct alternative to replace the blank line
		
		a) q=NULL; p->next=head; head =p ;
		b) q->next=NULL; head =p; p->next = head;
		c) head=p; p->next=q; q->next=NULL;
		d) q->next=NULL; p->next=head; head=p;
			CO3	d)

Q.No:3	**MCQ**	The postfix expression for the infix expression A+B*(C+D)/F+D*E a) AB+CD+*F/D+E* b) ABCD+*F/+DE*+ c) A*B+CD/F*DE++ d) A+*BCD/F*DE++	CO4	b)

	MCQ	The following postfix expression with single digit operands is evaluated using a stack 8, 2, 3,^,/,2, 3,*,+,5, 1,*,- The top two elements of the stack after the first * is evaluated a) 6,1 b) 5,7 c) 3,2 d) 1,5	CO4	a)
	MCQ	Assume that the operators +,-,* are left associative and ^ is right associative. The order of precedence (from highest to lowest) is ^,*,+,-. The postfix expression corresponding to the infix expression a+b*c-d^e^f is a)abc*+def^^- b)abc*+de^f^ c)ab+c*d-e^f^ d) -+a*bc^^def	CO4	a)
	MCQ	The following sequence of operation is performed on a stack: PUSH(10),PUSH(20),POP,PUSH(10), PUSH(20), POP, POP, POP, PUSH(20), POP The sequence of values popped out is: a) 20,10,20,10,20 b) 20,20,10,10,20 c) 10,20,20,10,20 d) 20,20,10,20,10	CO4	b)
Q.No:4	**MCQ**	The following numbers are inserted into an empty binary search-tree in the given order:10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree? a) 2 b) 3 c) 4 d) 5	CO4	b)
	MCQ	Postorder traversal of a given binary search tree, T produces the following sequence of keys 10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29 Which one of the following sequences of keys can be the result of an inorder traversal of the tree T? a) 9,10,15,22,23,25,27,29,40,50,60,95 b) 9,10,15,22,40,50,60,95,23,25,27,29 c) 29,15,9,10,25,22,23,27,40,60,50,95 d) 95,50,60,40,27,23,22,25,10,9,15,29	CO4	a)
	MCQ	What is the maximum height of 15 nodes? Assume that the height of a single node is 0. a) 2 b) 3 c) 4 d) 5	CO4	c)
	MCQ	The preorder traversal sequence of a binary search tree is 30,20,10,15,25,23,39,35,42.	CO4	a)

		Which of the following is the postorder traversal sequence of the same tree? a) 15,10,23,25,20,35,42,39,30 b) 15,20,10,23,25,42,35,39,30 c) 10,15,20,23,25,35,42,39,30 d) Cannot be determined		
Q.No:5	**MCQ**	A binary tree T has n leaf nodes. The number of nodes of degree 2 in T is a) log n b) n-1 c) n d) 2^n	CO4	b)
	MCQ	The number of leaf nodes in a rooted tree of n nodes, with each node having 0 or 3 children is a) n/2 b) (n-1)/2 c) (n-1)/3 d) (2n+1)/3	CO4	d)
	MCQ	The most efficient algorithm for finding the number of connected components in an undirected graph of n vertices and m edges has time complexity: a) O(n) b) O(m) c) O(m+n) d) O(m*n)	CO4	c)
	MCQ	Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time of Depth first Search on G, when G is represented as an adjacency matrix? a) θ(n) b) θ(n+m) c) $\theta(n^2)$ d) $\theta(m^2)$	CO4	c)
Q.No:6	**MCQ**	Given the following input (4322,1334,1471,9679,1989,6171,6173,4199) and the hash function x mod 10, which of the following statements are true? a) 9679,1989,4199 hash to the same value b) 1471,1334,4322 hash to the same value c) All elements hash to the same value d) Each element hashes to different value	CO5	a)
	MCQ	Consider a hash table with 9 slots. The hash function is h(k)=k mod 9 and the collisions are resolved by chaining . The following 9 keys are inserted in the order : 5, 28, 19, 15, 20,33,12, 17, 10. The maximum and minimum chain lengths in the hash table respectively are a) 3 and 0 b) 3 and 3 c) 4 and 0 d) 3 and 2	CO5	a)

	MCQ	Consider the hash table of size seven, with starting index zero, and a hash function (3x+4) mod 7. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1,3,8,10 is inserted into the table using closed hashing? Note that – denotes an empty location in the table a) 8,-,-,-,-,-,10 b) 1,8,10,-,-,-,3 c) 1,-,-,-,-,-,3 d) 1,10,8,-,-,-,3	CO5	b)
	MCQ	An advantage of chained hash table over the open addressing scheme is a) worst case complexity of search operation is less b) space used is less c) deletion is easier d) None of the above	CO5	c)
Q.No:7	**MCQ**	If one uses merge sort algorithm to sort the following elements in ascending order: 20,47,15,8,9,4,40,30,12,17 Then the order of these elements after second pass of the algorithm is a) 8,9,15,20,47,4,12,17,30,40 b) 8,15,20,47,4,9,30,40,12,17 c) 15,20,47,4,8,9,12,30,40,17 d) 4,8,9,15,20,47,12,17 e) 30,40	CO5	b)
	MCQ	Which of the following sorting algorithms has the lowest worst case complexity? a) Merge sort b) Bubble sort c) Quick sort d) Selection sort	CO5	a)
	MCQ	For merging two sorted lists of size m and n into a sorted list of size m+n, the number of comparisons required a) O(m) b) On) c) O(m+n) d) O(m*n)	CO5	c)
	MCQ	The worst case time complexity of quick sort is a) O(n^2) b) O(nlogn) c) O(n) d) O(n^3)	CO5	a)

SECTION-B(Answer Any Three Questions. Each Question carries 12 Marks)

Time: 1 Hour and 30 Minutes **(3×12=36 Marks)**

Question No	Question	CO Mapping (Each question should be from the same CO(s))
Q.No:8	a) What are the disadvantages of array and how can we overcome that? [4] b) Given a 2-D array A[-100:100,-5:50]. Find the address of element A[99,49] considering the base address 100 and each element requires 4 bytes of storage. Follow row major order. [2] c) Given an integer array, find the contiguous subarray (containing at least one number) which has the largest sum and return its sum? [6]	CO1,3
	What is sparse matrix? How to represent sparse matrix in 3-tuple format? Write the pseudo code to add two sparse matrices.	
	a)Write an algorithm /Pseudo code/C code snippet to remove duplicate elements from unsorted single linked list. [6] b) Write an algorithm /Pseudo code to merge two sorted linked list to produce a sorted linked list.[6]	
Q.No:9	a) Convert the following infix expression to postfix expression using Stack. [6] a+b/(c^d-e)^(f+g*h)-i*j*k*l b) Evaluate the following postfix expression using stack. 12,7,3,-,/,2,1,5,+,*,+ [6]	CO4,5
	a) What is Deque? Write algorithm to implement an input restricted deque. [8] b) Explain how to implement two stacks in one array A[1…n] in an efficient manner. [4]	
	Write the pseudo code for Merge sort including merge function. Also illustrate the step-by-step process with a given array: {10, 80, 30, 90, 40, 50, 70}	
Q.No:10	a) Construct a binary search tree for the following sequence of numbers. [6] 45,36,76,23,89,115,98,39,41,56,69,48 b) What is an AVL tree? Construct an AVL tree with the	CO4

	following set of numbers. [6] 55,66,77,15,11,33,22,35,25	
	a) What is B-tree ? Difference between B tree and B+ tree . [4] b) Construct a max heap tree and a min heap tree with the following data elements. [8] 25, 30, 15, 8, 27, 22, 12, 7, 18	
	What is the use of AVL trees ? Discuss all rotations to ensure that the tree remains balanced after every insertion. Illustrate the steps involved while building an AVL tree with values:15,20,24,10,13,7,30,36,25.	
Q.No:11	a) What do you mean by collision in hashing ? [2] b) What is chaining and write its advantages? [2] c) The following values are to be stored in a hash table 25,42,96,101,102,162,197 Describe how the values are hashed by using division method of hashing with a table size of 7. Use chaining as the method of collision resolution. [8]	CO_5,6
	a) What do you mean by collision in hashing and how to handle the collisions? [6] b) Draw the neat diagram to illustrate separate chaining method to handle collisions using hash function as “key mod 7”and sequence of keys as 50,700.76,85,92,73,101. [6]	
	Discuss both the techniques of BFS and DFS with suitable data structure and justify the selection of data structure.	